BRAIN TUMOR CLASSIFICATION USING DEEP LEARNING IMAGE PROCESSING ON MRI IMAGES

CAPSTONE PROJECT REPORT

Submitted by

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DECLARATION

We affirm that the project work titled **“ Brain Tumor Classification Using Deep Learning Image Processing on MRI Images”** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed the part of any other project work submitted for award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

Certified that this project report **"Brain Tumor Classification Using Deep Learning Image Processing on MRI Images"** is the bonafide work of "**SK.RASOOL REHMAN,Y.SUMANTH KUMAR, T.PARDHASARADHI REDDY"** who carried out the project work under my supervision.

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Internal Examinar

External Examinar

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Project Summary

Project Title	Brain Tumor Classification Using Deep Learning Image Processing on MRI Images
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Guide Name/Designation	Dr.N.V.S.Natteshan, Associate Professor, Department of Computer Science and Engineering
Program Concentration Area	Medical Image Classification
Technical Requirements	Jupyter tool is used by the developer to complete the project.

Engineering standards and realistic constraints in these areas: (Refer Appendix in page 4 of this doc.)

Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic		
Environmental		
Social		

Ethical		
Health and Safety	This project is mainly used to support the radiologist for better identification of tumor using neural networks	✓
Manufacturability		
Sustainability		

Realistic Constraints:

Health and Safety:

In the research of medical, various segmentation methods have been proposed to identify the lesions in the beginning stage to save the millions of human being. Still it is challenging for find out the complex tumors present in the MR brain image. Deep learning neural networks is used to analyze the various complex tumors in deeply. The main focus of this project is to locate the various tumors present in the magnetic resonance (MR) brain image using deep learning neural networks. Because of multifaceted structure of brain, better examination and study is required by a radiologist to identify the various tumors. With the support of neural networks identification of the various tumors is effectively performed. These processes support the radiologist extensively to perform better diagnosis identifying the various types of tumors in the early stages.

Engineering standards:

This project is based on IEEE 3333.2.1-2015 –Medical modeling and visualization

This standard focuses on the project demands arising when scientific results in the field of medical visualization are applied for the construction of a software system. It is targeted to aid the clinical work of medical professionals. This standard includes visualization techniques by the automated medical shape detection and reconstruction of threedimensional (3D)

models from two-dimensional (2D) medical images. When the MR image is given as input, automatically the tumors and tissue portion will be demarcated separately. Finally the region of lesions are segmented by the optimization techniques are compared with the gold standard image obtained by the radiologist to check the efficiency of the suggested methodologies and the segmented results can be reconstructed for further analysis to get better visualization of tumors.

ABSTRACT

Brain tumor classification is a critical task in medical image analysis, enabling accurate diagnosis, treatment planning and monitoring of patients with brain tumors. Using magnetic resonance imaging (MRI),In order to detect and classify brain cancers, deep learning is used to build models. This brain cancer can be easily and quickly identified because of this. primarily the cause of brain problems uncontrolled brain cell growth, which can deteriorate the brain's structural integrity and ultimately lead to brain cancer that is aggressive. The early detection of brain tumors and the subsequent effective treatment of getting treated could reduce the death rate. Precise and effective brain tumor segmentation plays a pivotal role in enabling clinicians and radiologists to evaluate essential factors such as tumor dimensions, spatial location, and developmental status. In this article, we propose a convolutional neural network (CNN) architecture: VGG16, VGG19, ResNet-50, ResNet-101, MobileNetV2, Inception V3, and EfficientNetB0 for accurate brain tumor detection from MRI images.

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LIST OF ABBREVIATIONS

FIGURES	DETAILS
CNN	Convolutional Neural Network
VGG	Visual Geometry Group
Resnet	Residual Neural Network
MRI	Magenetic Resonance Imaging

CHAPTER- I

INTRODUCTION

1.1 Overview

A malignancy that can have a terrible effect on a person's life is a brain tumor. Early detection is crucial for the condition to be successfully treated, and this has proven to be a constant problem for medical personnel. Thanks to recent developments in computer vision and machine learning, convolutional neural networks (CNNs) can now be used to precisely detect and categorize brain cancers. We will cover the many methods and approaches utilized to accomplish this in this study as we discuss how CNNs can be used to detect brain cancers. Due to poor diagnosis of diseases important resources are wasted daily in the world's health care industry. Miss diagnosis of any disease may lead to unnecessary additional medical tests, may delay in treatment schedules and may affect the survival of the patient.

Now-a-days, many health care and technology innovators are experimenting on Artificial Intelligence and Machine Learning. In the recent period, there is a significant advancement in the medical field due to artificial intelligence, image processing and deep learning techniques which in turn help in faster and easy diagnosis. Otherwise it takes a lot of time. This makes it quite simple for the doctors to resolve the difficult instance that takes a lot of time. We required highly effective and trustworthy computer aided technology to diagnose life threatening diseases like cancer which is a major contributor to the global morality rate. In order to overcome such restrictions.

The human brain is a complex and vital organ responsible for various cognitive functions and bodily control. However, it is susceptible to a range of abnormalities, one of the most concerning being brain tumors. Brain tumors can be benign or malignant, and their early detection and accurate diagnosis are crucial for effective treatment and improved patient outcomes.

Medical imaging, especially Magnetic Resonance Imaging (MRI), has become a cornerstone in the diagnosis of brain tumors due to its non-invasive nature and high-resolution images. However, the manual interpretation of these images by radiologists is time-consuming and can be prone to human error. In 2020, brain tumor was ranked as the 10th most common kind of tumor among Indians. The International Association of Cancer Registries(IARC) reported that there are over 28,000 cases of brain tumors reported in India each time and further than 24,000 people reportedly die due to brain tumor annually. In India, the frequency of brain tumors is steadily rising.

More and more cases of brain tumors are reported each year in our country among people of varied age

groups. Every time 40,000 to 50,000 people are diagnosed with brain tumors, out of which 20 percent are children. In India, the average rate of brain tumors is between 5 and 10 per 100,000 people, with a rising tendency, and they make up 2% of all malignancies.

1.2 CNN

CNNs are a class of deep learning algorithms that have demonstrated remarkable success in image analysis tasks. They have the ability to automatically learn and extract intricate patterns and features from medical images, such as Magnetic Resonance Imaging (MRI) scans.

The primary goal of brain tumor classification using CNN algorithms is to develop a computer-aided diagnostic system that can assist radiologists and healthcare professionals in accurately identifying and categorizing brain tumors from medical imaging data. This not only reduces the burden on medical practitioners but also enhances diagnostic accuracy, leading to more effective treatment planning and improved patient outcomes.

1.3 Fundamental Steps

1. Data Collection 2. Data Preprocessing (Train-Test Split) 3. CNN Architecture Design(Convolutional Layers,Pooling Layers,Fully Connected Layers) 4. Model Training(Backpropagation,Optimization) 5. Model Evaluation(Accuracy,Precision,Recall,F1 Score) 6. Fine-tuning(Optional) 7. Prediction(New Brain Images) 8. Post-processing(Optional) 9. Results Analysis 10. Documentation & Report

1.4 Feature Extraction

1. **Convolutional operation:** A convolutional layer is used to compress an image. Using convolutional operation, a feature map is produced. The feature map also makes processing the image simpler and quicker. Each image has unique qualities. Convolutional operations use feature detectors to identify a specific feature of an image. This is how many filters the convolutional layers will use to learn.

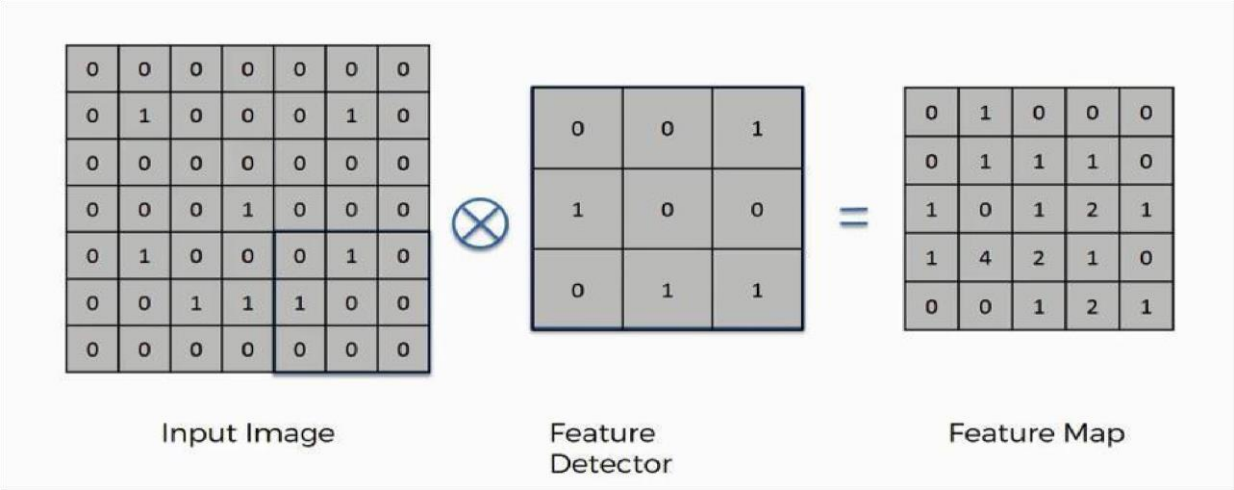


Fig-1 Convolutional operation

2. Maxpooling: A complicated and large-sized MRI image must be reduced while keeping all of the image's significant details and patterns. The "max pooling" method of pooling selects the most components possible from the feature map area that the filter covers. After maximal pooling, only the most important elements of the feature map will be left.

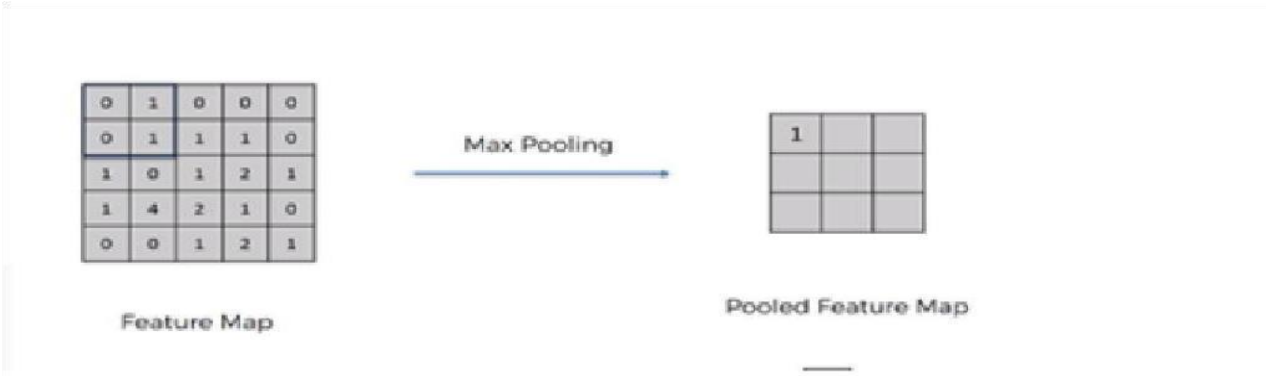


Fig-2 Maxpooling

Through study in the filed, then used the brain MRI images and produce a model for predicting tumorous and non-tumorous images of the given MRI images using data augmentation technique and convolutional neural network.

CHAPTER- II

LITERATURE REVIEW

In this part, we reviewed earlier studies that looked at ML models for detecting brain tumors. We have listed a few earlier efforts in Table 1. A deep educational model employing an MRI dataset for brain tumor diagnosis was proposed by Almadhoun et al.

[1]. In addition to the deep educational model, they used the VGG16, MobileNet, ResNet-50, and Inception V3 transfer learning models. To test their models, they employed a dataset of 10,000 MRI images with a resolution of 200x 200 pixels. Brain tumors and other tumors were separated into two categories, each containing 5000 photos. The deep educational model they developed performed better; training accuracy was 100% and test accuracy was 98%. In order to identify MRI pictures of brain tumors, Nayak et al.

[2] introduced dense EfficientNet, a CNN-based network. Additionally, ResNet-50, MobileNet, and MobileNetV2 were examined by the researchers; they found that their dense EfficientNet outperformed them all. After training the dense EfficientNet model, they achieved a 98.78% accuracy and a 98.0% F1-score. Their study used four different MRI scans to spot brain tumors. There were 3260 MR pictures in the entire dataset.

Using an MRI dataset to identify brain tumors, Musallam et al.

[3] created a DCNN model. Iterations, maxpooling, and a light model with a few convolutions were employed in their approach. VGG16, VGG19, and CNNSVM were all examined by the researchers. The four subcategories of the 3394 MRI scans were: no tumor (606), meningioma (945), glioma (934), and pituitary (909). The proposed model had a detection rate of 99% for gliomas, 98.26% for meningiomas, 95.95% for pituitary, and 97.14% for normal pictures, yielding an overall accuracy of 97.72%. A CNN-based residual network for

the early diagnosis of brain cancers was introduced by Obeidavi et al.

[4] utilizing a dataset of 2000 MR images. The BRATS 2015 MRI dataset was used, and the results for the residual networks were encouraging. Their suggested model has a 97.05% accuracy rate. They also took into account additional criteria, and as a result, their average accuracy was 97.05%, their average global accuracy was 94.43%, their average IoU was 54.21%, their average weighted IoU was 93.64%, and their average BF score was 57.027%. During training, 100 epochs were used to enhance performance. For deep neural network topologies, Wozniak et al.

[5] created a state-of-the-art correlation learning method (CLM) that combines a CNN with a traditional architecture. The 3064 brain tumors they looked at included meningiomas (708 photos), gliomas (1426 images), and pituitary tumors (930 images). Their created CLM model exhibited a recall of almost 95%, a precision of roughly 95%, and an accuracy of roughly 96%. For the purpose of identifying brain tumors, Garg et al.

[6] proposed the naive Bayes, random forest, neural network, KNN, and decision tree machine learning models, as well as a hybrid ensemble classifier (KNN-RF-DT). With 85% of the data used for training and 15% used for testing, they used 2556 pictures of brain tumors to evaluate the machine learning models. Thirteen features were found for the categorization as a consequence of feature extraction by SWT, PCA, and GLCM. The proposed method for recognizing and classifying brain tumors was assessed, and the findings revealed that the technique has 97.305% accuracy, 97.73% precision, 97.60% specificity, 97.04% sensitivity, and 97.41% reliability.

Models	Dataset	Accuracy
[1] Deep educational	10,000 MRI Images	98%
[2] Dense EfficientNet	3260 MRI Images	98.78%
[3] DCNN	3394 MRI Images	97.72%
[4] Residual network	BRATS 2015 datasets, 2000 MRI Images	97.04%
[5] CLM with CNN	3064, CT brain scan images	96%

[6] Hybrid ensemble classifier	2556 brain tumor images	97.305%

Table-1 Model Survey

CHAPTER- III

PROBLEM STATEMENT

Accurate and timely diagnosis of brain tumors is crucial for effective treatment planning and improved patient outcomes. However, the manual interpretation of medical imaging data, such as magnetic resonance imaging (MRI) scans, is time-consuming and subject to human error. To address this challenge, we aim to develop an automated brain tumor classification system that can detect the presence of tumors and classify them into specific types or subtypes.

The key challenges include handling variations in tumor size, location, and appearance, as well as achieving high accuracy and interpretability to aid medical decision-making.

CHAPTER- IV

REQUIREMENTS

4.1 Requirement Description

Dataset for this project is a diverse and comprehensive dataset of brain MRI images with associated labels, including information on whether the tumor is benign, malignant, or specific tumor types (e.g., glioma, meningioma, pituitary tumor). High-quality images with consistent resolution, contrast, and noise levels. The dataset should be free from artifacts and contain annotated regions of interest (ROIs) highlighting the tumors.

For the purpose of identifying brain tumors, we acquired the dataset from freely available open data on

kaggle. The dataset was developed using pictures from magnetic resonance imaging (MRI). We chose MRI pictures for our research since MRI is the finest method for discovering brain cancers. Meningioma tumor (937 images), no tumor (500 images), pituitary tumor (900 images), and glioma tumor (927 images) were the four distinct forms of brain tumors. information that was utilized in our investigation. In our dataset, 3264 MRI images are used. Figure1 exhibits the dataset's breakdown and the MRI images in accordance with brain tumors of many kinds.

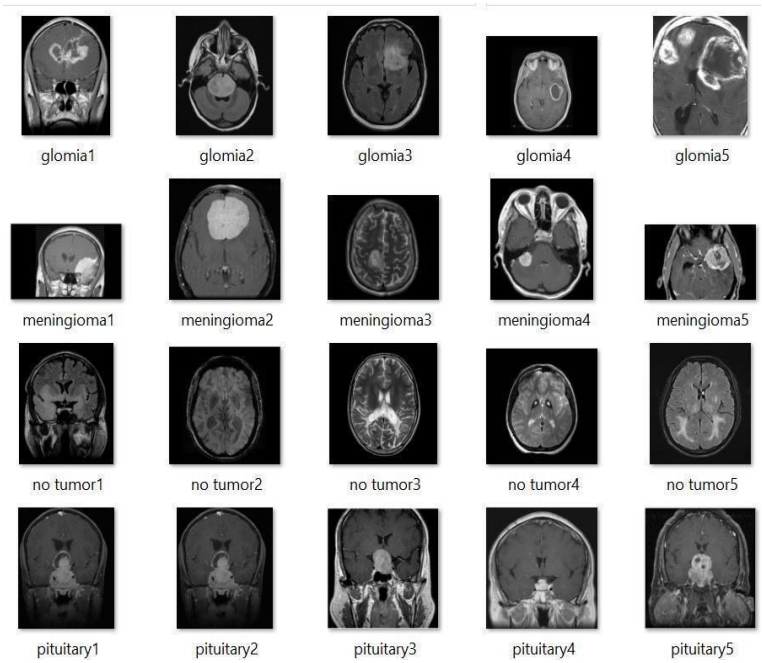
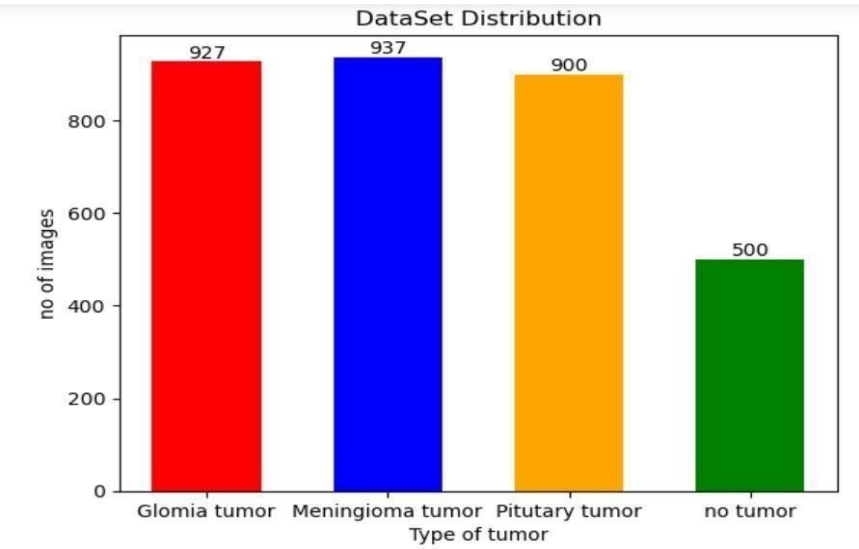


Fig-3 Sample image for Input dataset

Pre-processing, where the data are transformed to make them useful for training purposes, is a crucial stage. The MRI images were blurry and of low quality because they were taken from a patient database. At this point, we normalized our photos to get them ready for additional processing. We apply filters to the images in order to smooth them out and eliminate any blurry portions of the original images.



Graph-1 Describing about the data set distribution

4.2 Software Requirement

Jupyter Notebook is an open-source web application that allows you to create and share documents that contain live code, equations, visualizations, and narrative text. It is widely used in data science, machine learning, and scientific computing for interactive data analysis and exploration.

HTML AND CSS: Using HTML and CSS in the context of brain tumor classification or segmentation typically involves creating a web-based user interface or visualization tool for displaying medical images and the results of image analysis.

CHAPTER- V

SYSTEM DESIGN

To design a workflow specifically for training and implementing a ResNet (Residual Network) algorithm for brain tumor classification, We can follow a more detailed procedure tailored to the ResNet architecture. ResNet is a popular deep learning architecture known for its effectiveness in image classification tasks.

5.1 Workflow Diagram

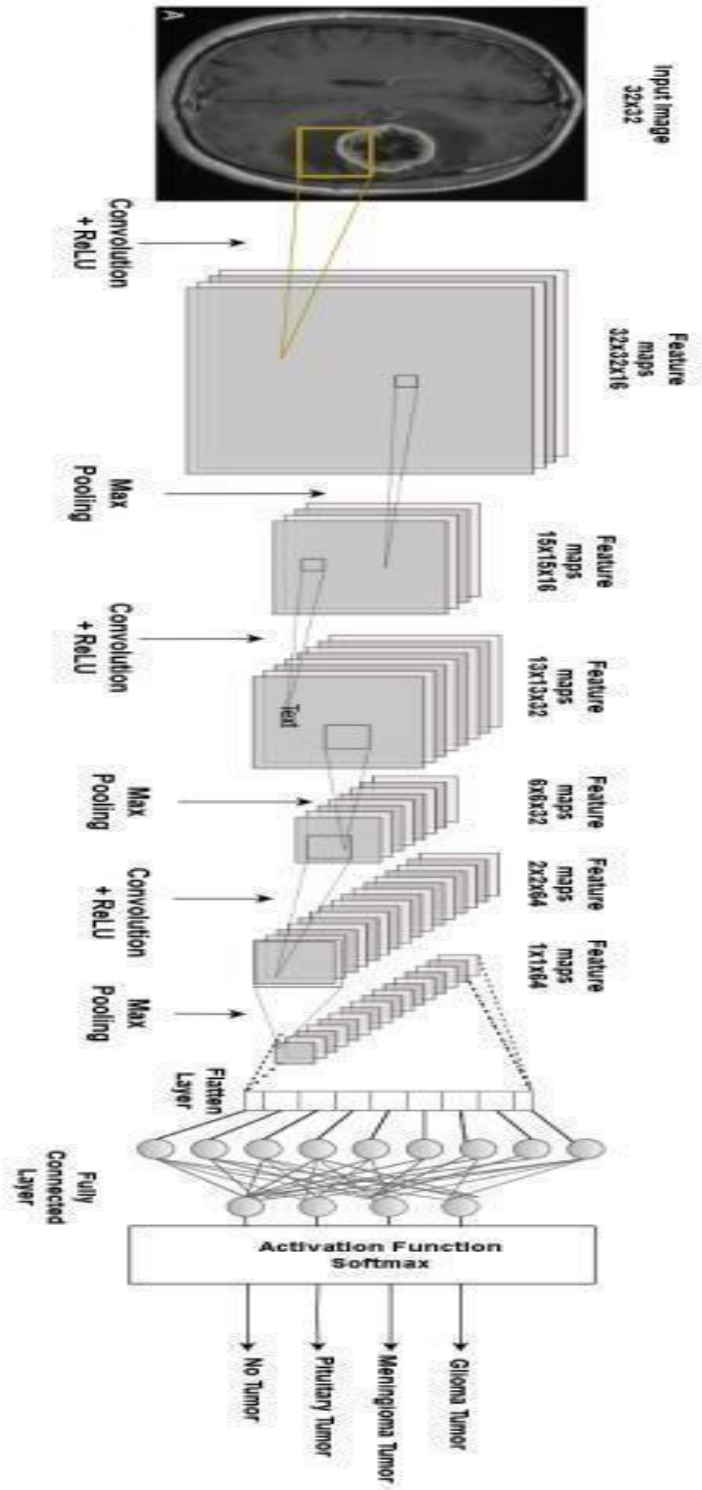


Fig-4 Proposed model

The process is divided into a few key phases. We preprocessed our datasets after gathering our data from a readily accessible web source. In the validation stage, we applied the holdout validation approach. We trained our images using a variety of machine learning algorithms. Three groups of our dataset were created: training, testing, and validation. Four different types of brain images: glioma tumors, meningioma tumors, no tumor, and pituitary tumors were subjected to validation. After that, we looked at a variety of parameters, including accuracy, loss, validation accuracy, and validation loss, in order to confirm our findings.

CHAPTER- VI

DESIGN ALTERNATIVES

6.1 Image Classification

Design alternatives for brain tumor classification in the context of image classification can be approached in several ways

1. **Traditional Machine Learning Models:** Instead of deep learning models like Convolutional Neural Networks (CNNs), you can use traditional machine learning models like Support Vector Machines (SVM), Random Forests, or K-Nearest Neighbors (K-NN) for image classification. Extract relevant features from the images (e.g., texture, shape, intensity) and use these as input to the model.
2. **Ensemble Models:** Combine multiple machine learning models or deep learning models, such as a combination of CNNs with SVMs or decision trees, to improve classification accuracy and robustness.
3. **Transfer Learning Variants:** Utilize transfer learning with different pre-trained models like Inception, VGG, or MobileNet instead of ResNet. Experiment with different model architectures to see which one works best for your specific dataset.

6.2 Image Segementation

Design alternatives for brain tumor segmentation, which involves delineating the boundaries of brain tumors within medical images, can be approached using various techniques and methods.

1. **Machine Learning-Based Segmentation**: Train machine learning models, such as Random Forests, Support Vector Machines, or k-means clustering, to segment brain tumor regions based on extracted features.
2. **Convolutional Neural Networks (CNNs):** Utilize deep learning models, like U-Net, Fully Convolutional Networks (FCNs), or Mask R-CNN, specifically designed for semantic and instance segmentation tasks. These models can be trained end-to-end to produce pixel-wise tumor segmentations. 3. **Transfer Learning with Pre-trained CNNs:** Fine-tune pre-trained CNN models on brain tumor segmentation tasks. Models like VGG, ResNet, and Inception can be adapted for this purpose.

CHAPTER- VII

PROPOSED APPROACH

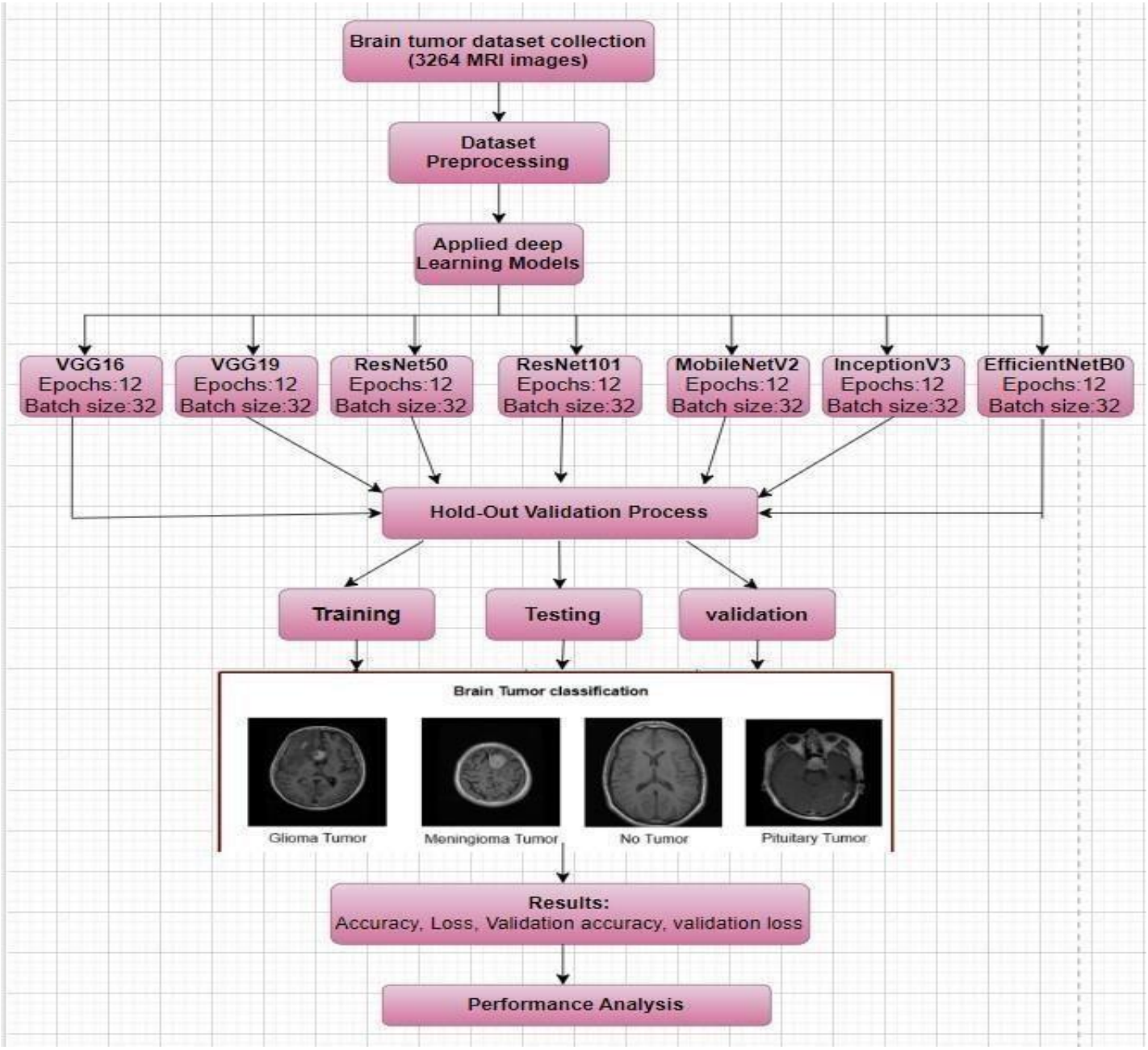


Fig-5 APPROACH

Model Training:

We divide the pixel values by 255 to normalize them before putting the photos into the model. In Input Layer that takes an input image of size 256x256 pixels with 3 color channels (RGB). On the basis of the training data, we train the model for a predetermined number of epochs, such as 12 and batch size 32 . We track the loss and accuracy metrics during training to evaluate the model's effectiveness.

Model Evaluation :

We evaluate the model's performance after training using the testing dataset, which it has never seen before. This step assesses the model's adaptation to new, untested tasks. We compute typical performance metrics such as loss and accuracy to evaluate the model's performance. These metrics provide information on how well the model categorizes the brain tumor.

We use transfer learning to identify whether a tumor is the type of brain tumor among glioma, pitutary, meningioma and no tumors. The model selection for recognising a tumor is a very important aspect and to get better results. Seven deep learning pre-trained models are used to recognize brain tumors, namely VGG16, VGG19, RESNET50, RESNET101, MOBILENETV2, INCEPTIONNETV3, AND EFFICIENTNETB0.

7.1 Algorithms Used

1. **VGG16:** VGG16 is a convolutional neural network with 16 layers. It comprises 13 convolutional and 3 fully connected layers, employing a stride of 1, and incorporating max-pooling layers to reduce spatial dimensions. In convolution layers ReLU activation function is used to filter the images and softmax activation function is used in fully connected layers to detect the brain tumor. VGG19 is the extension of VGG16 model with 16 convolutional layers and 3 fully connected layers .It is a pretrained model.
2. **ResNet-50:** ResNet-50 is a pretrained Deep Learning model for image classification based on the Convolutional Neural Network (CNN), a type of deep neural network that is most typically used to analyze visual data.
3. **ResNet101:**ResNet101 has 104 convolutional layers made up of 33 blocks of layers, 29 of which are directly used in previous blocks. ResNet model pre-trained on ImageNet-1k at 224x224 resolution. ResNet is an artificial neural network with a "identity shortcut connection," which allows the model to skip one or more levels. This method allows the network to be trained on hundreds of layers without affecting performance.
4. **MobileNetV2:**MobileNetV2 is a convolutional neural network with 48 layers. It reduce the number of parameters and computations where memory and processing power is limited.
5. **InceptionV3:**InceptionV3 is a convolutional neural network with 53 layers. It uses Inception series that consists of multiple parallel convolutional layers of different kernal sizes which extracts features at various scales.
6. **EfficientNetB0:**EfficientNetB0 can train more than one millon images and classify the images of different classes. The above classification models are trained and validated. The best model is chosen

among them to predict the accurate results whether the person having which type of tumor among glioma, pitutary, meningioma and no tumors.

7.2 Approaching Method

RESNET50: ResNet-50 is a deep convolutional neural network (CNN) architecture that can be used for various computer vision tasks, including brain tumor classification and segmentation. It's a variant of the ResNet (Residual Network) family of models and is particularly well-suited for tasks that require the recognition of intricate patterns and features in images.

Here's how ResNet-50 can work in the context of brain tumor classification or segmentation:

1. **Feature Learning:** ResNet-50 is trained on a large dataset of diverse images (e.g., ImageNet) to learn general image features, which are represented in the earlier layers of the network. These features include basic shapes, edges, and textures.
2. **Transfer Learning:** After pre-training on a large dataset, the ResNet-50 model can be finetuned for brain tumor classification or segmentation using a smaller, domain-specific dataset of medical images. This approach is known as transfer learning.
3. **Image Classification:** For brain tumor classification, you can modify the output layer of ResNet50 to have as many neurons as there are tumor classes (e.g., benign, malignant). The model can then classify an input image into one of these classes based on the patterns it has learned.
4. **Tumor Segmentation:** ResNet-50 can also be used for semantic segmentation, where it classifies each pixel in an image as either tumor or non-tumor. You can modify the network's architecture to include an output layer that produces a mask representing the tumor region.
5. **Interpretable Features:** The deep layers of ResNet-50 can capture more abstract features, such as the texture and spatial relationships of tumors, which can be useful for tasks like distinguishing between tumor subtypes.
6. **High Accuracy:** ResNet-50, with its deep architecture, has the potential to achieve high accuracy in brain tumor classification and segmentation tasks when fine-tuned properly, especially if you have a sufficiently large and representative dataset.

7.3 Deployment

FLASK: Deploying a web-based brain tumor classification or segmentation tool using Flask involves setting up a server to host the HTML and CSS user interface, as well as the backend processing system. Flask is a lightweight web framework for Python that is well-suited for creating web applications.Here are the steps to deploy the model

1. Install Flask
2. Organize Your Project
3. Create Flask Application
4. HTML Templates and CSS Styling
5. Backend Processing
6. Deploy with a Production Web Server

Flask is a flexible and straightforward framework for deploying web applications, and it can be a valuable tool for creating web-based interfaces for brain tumor classification or segmentation tools.

CHAPTER- VIII

IMPLEMENTATION AND RESULT

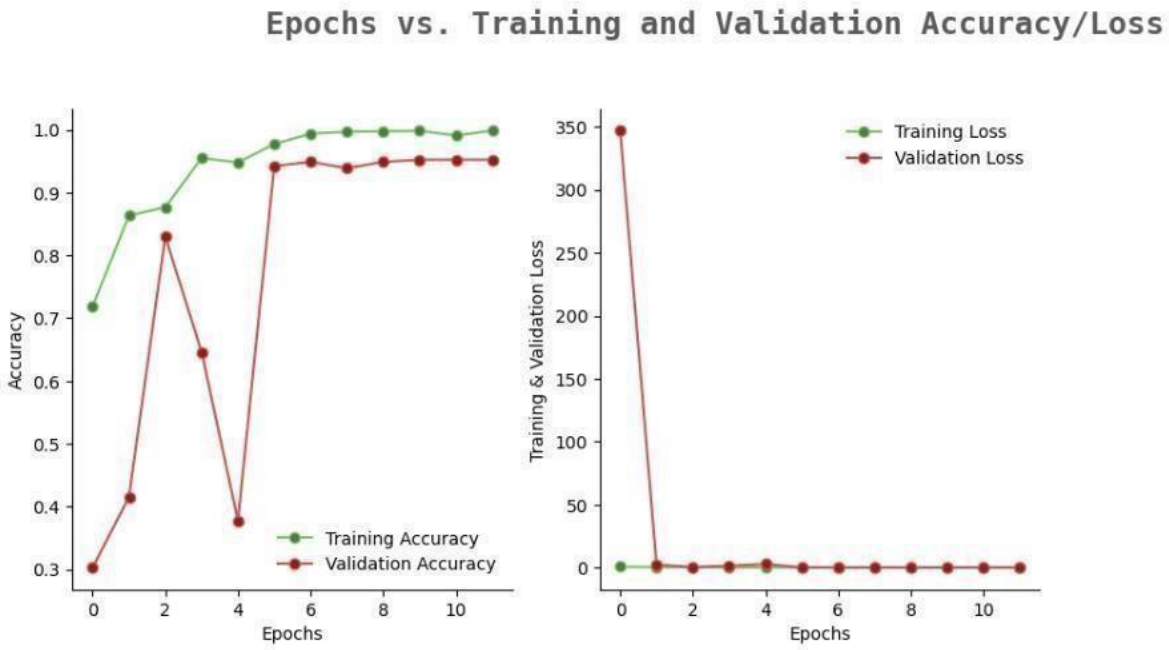
8.1 Coding

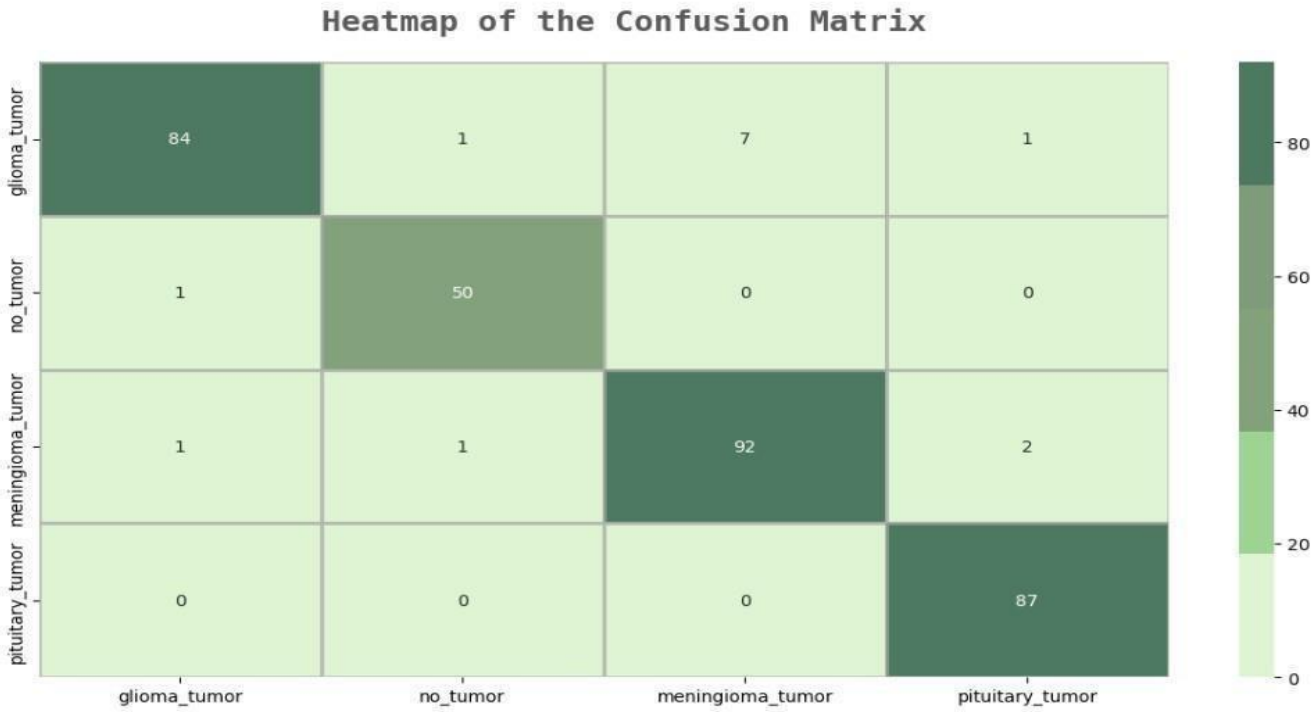
```
import matplotlib.pyplot as plt  import numpy as np import pandas as pd import
seaborn as sns import cv2  import tensorflow as tf  from
tensorflow.keras.preprocessing.image import ImageDataGenerator from tqdm import
tqdm  import os  from sklearn.utils import shuffle from  sklearn.model_selection
import train_test_split from tensorflow.keras.applications import ResNet50
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau, TensorBoard,
ModelCheckpoint from sklearn.metrics import classification_report,confusion_matrix import ipywidgets as
widgets import io  from PIL import Image  from IPython.display import display,clear_output from warnings
import filterwarnings for dirname, _, filenames in os.walk("brain tumor dataset"):     for filename in
filenames:        print(os.path.join(dirname, filename))   colors_dark = ["#1F1F1F", "#313131", '#636363',
'#AEAEAE', '#DADADA'] colors_red = ["#331313", "#582626", '#9E1717', '#D35151', '#E9B4B4']
colors_green = ['#01411C','#4B6F44','#4F7942','#74C365','#D0F0C0']

sns.palplot(colors_dark) sns.palplot(colors_green) sns.palplot(colors_red)  labels
= ['glioma_tumor','no_tumor','meningioma_tumor','pituitary_tumor']  X_train = []
y_train = [] image_size = 150 for i in labels:      folderPath = os.path.join("brain
          tumor  dataset","Training",i)                                    for       j
          in tqdm(os.listdir(folderPath)):
                                    img      = cv2.imread(os.path.join(folderPath,j))
                                                                                  img      =
cv2.resize(img,(image_size, image_size))         X_train.append(img)        y_train.append(i)


for i in labels:
    folderPath = os.path.join("brain tumor dataset","Testing",i)     for
j in tqdm(os.listdir(folderPath)):        img =
cv2.imread(os.path.join(folderPath,j))        img =
cv2.resize(img,(image_size,image_size))
X_train.append(img)        y_train.append(i)
```

```
X_train = np.array(X_train) y_train = np.array(y_train)   k=0   fig, ax = plt.subplots(1,4,figsize=(20,20))
fig.text(s='Sample Image From Each
Label',size=18,fontweight='bold',
fontname='monospace',color=colors_dark[1],y=0.62,x=0.4,alpha=0.8) for i in labels:     j=0
while True :        if y_train[j]==i:              ax[k].imshow(X_train[j])
ax[k].set_title(y_train[j])            ax[k].axis('off')            k+=1            break        j+=1
X_train, y_train = shuffle(X_train,y_train, random_state=101) X_train.shape
X_train,X_test,y_train,y_test = train_test_split(X_train,y_train, test_size=0.1,random_state=101) y_train_new
= [] for i in y_train:
   y_train_new.append(labels.index(i)) y_train = y_train_new   y_train =
tf.keras.utils.to_categorical(y_train)


y_test_new = [] for i in y_test:
y_test_new.append(labels.index(i)) y_test = y_test_new    y_test
= tf.keras.utils.to_categorical(y_test)
effnet = ResNet50(weights='imagenet',include_top=False,input_shape=(image_size,image_size,3)) model =
effnet.output
model = tf.keras.layers.GlobalAveragePooling2D()(model) model = tf.keras.layers.Dropout(rate=0.5)(model)
model = tf.keras.layers.Dense(4,activation='softmax')(model) model =
tf.keras.models.Model(inputs=effnet.input, outputs = model) model.summary()
model.compile(loss='categorical_crossentropy',optimizer = 'Adam', metrics= ['accuracy']) tensorboard =
TensorBoard(log_dir = 'logs') checkpoint =
ModelCheckpoint("effnet.h5",monitor="val_accuracy",save_best_only=True,mode="auto",verbose=1)
reduce_lr = ReduceLROnPlateau(monitor = 'val_accuracy', factor = 0.3, patience = 2, min_delta =
0.001,mode='auto',verbose=1) import tensorflow as tf class
myCallback(tf.keras.callbacks.Callback):    def on_epoch_end(self,epoch,logs={}):
print("call")        if(logs.get('accuracy')>.99):
        print("\nReached %2.2f%% accuracy,so stopping training !!" %(99))
self.model.stop_training=True                callbacks=myCallback()                history          =
model.fit(X_train,y_train,validation_split=0.1,        epochs          =12,    verbose=1,
batch_size=32,callbacks=[tensorboard,checkpoint,reduce_lr])
```

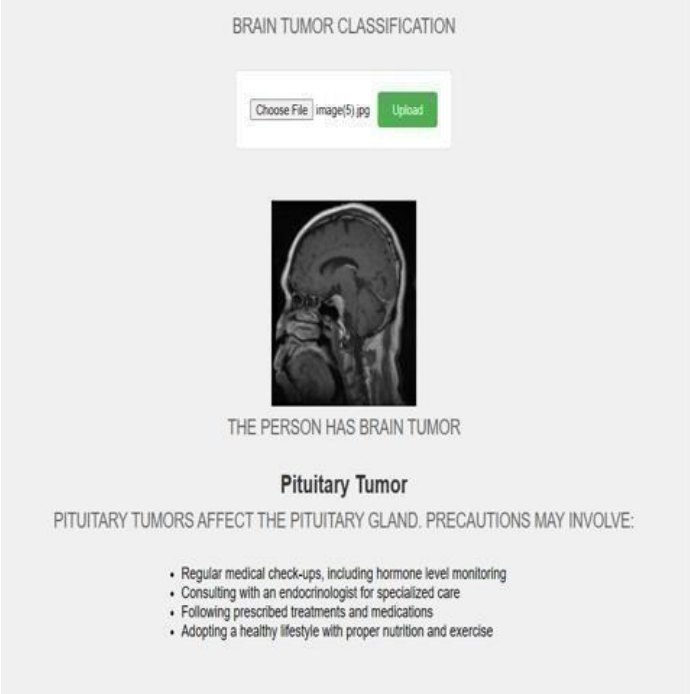
```
filterwarnings('ignore')

epochs = [i for i in range(12)]  fig, ax =
plt.subplots(1,2,figsize=(10,5)) train_acc =
history.history['accuracy'] train_loss =
history.history['loss'] val_acc
= history.history['val_accuracy'] val_loss
= history.history['val_loss']

fig.text(s='Epochs vs. Training and Validation Accuracy/Loss',size=18,fontweight='bold',
fontname='monospace',color=colors_dark[1],y=1,x=0.28,alpha=0.8)

sns.despine()   ax[0].plot(epochs, train_acc,
marker='o',markerfacecolor=colors_green[2],color=colors_green[3],          label
= 'Training Accuracy')   ax[0].plot(epochs, val_acc,
marker='o',markerfacecolor=colors_red[2],color=colors_red[3],          label  =
'Validation Accuracy') ax[0].legend(frameon=False) ax[0].set_xlabel('Epochs')
ax[0].set_ylabel('Accuracy')

sns.despine()   ax[1].plot(epochs, train_loss,
marker='o',markerfacecolor=colors_green[2],color=colors_green[3],          label
='Training Loss')   ax[1].plot(epochs, val_loss,
marker='o',markerfacecolor=colors_red[2],color=colors_red[3],          label = 'Validation
Loss') ax[1].legend(frameon=False) ax[1].set_xlabel('Epochs')  ax[1].set_ylabel('Training &
Validation Loss')

fig.show() pred = model.predict(X_test) pred
= np.argmax(pred,axis=1) y_test_new
=                np.argmax(y_test,axis=1)                          print(classification_report(y_test_new,pred))
fig,ax=plt.subplots(1,1,figsize=(14,7))
sns.heatmap(confusion_matrix(y_test_new,pred),ax=ax,xticklabels=labels,yticklabels=labels,annot=True,
cmap=colors_green[::-1],alpha=0.7,linewidths=2,linecolor=colors_dark[3])   fig.text(s='Heatmap  of  the
Confusion Matrix',size=18,fontweight='bold',
fontname='monospace',color=colors_dark[1],y=0.92,x=0.28,alpha=0.8)

plt.show()
model.save('braintumor')
```

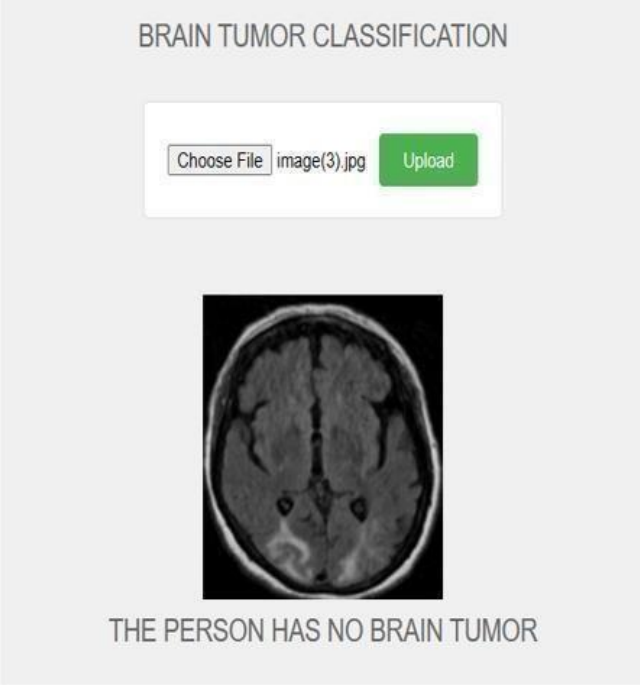
8.2 Output





Graph-2 Epochs vs Train and val_accuracy/loss





Graph-3 Confusion Matrix

Fig-6 Pituitary _tumor

Fig-7 No_tumor

Fig-7 Glioma_tumor

Fig-8 Meningioma_tumor

8.3 Result

The results of several types of constructed deep learning models—namely, the VGG16, VGG19, ResNet50, ResNet101 MobileNetV2, Inception V3, and EfficientNetB0 classification algorithms—on the brain

tumor MRI image dataset are examined in Table 2, and comparisons are illustrated in Figure 5. Table 2 shows the model performance in terms of accuracy, loss, validation accuracy, and validation loss function outcomes. Based on the findings in Table 1, it was discovered that the ResNet101 beat the other deep learning models after assessing the approaches of the CNN, namely, VGG16, VGG19 CNN, ResNet-50, ResNet101 MobileNetV2, Inception V3, and EfficientNetB0. The ResNet101 has a training accuracy of 99.89%, a training loss of 0.0051, a validation accuracy of 97.2%, and a validation loss of 0.083.

Model	**Accuracy**	**loss**	**Val_accuracy**	**Val_loss**
VGG16	81.04%	0.5104	81.63%	0.4686
VGG19	66.86%	0.6686	65.65%	0.7374
ResNet50	99.89%	0.0031	95.92%	0.1378
ResNet101	99.89%	0.0051	97.28%	0.0832
MobileNetV2	99.8%	0.0063	88.78%	0.8878
InceptionV3	99.21%	0.022	95.92%	0.1261
EfficientNetB0	99.81%	0.0057	96.94%	0.0883

Table-2 models vs accuracy

CHAPTER- IX

IMPLEMENTATION AND RESULT

9.1 Conclusion

Early identification of brain tumors has the potential to significantly reduce worldwide mortality rates. Because of the tumor's shape, changing size, and structure, detecting tumors in the brain remains difficult. The classification of MRI images has a significant impact on clinical diagnosis and treatment selection for patients with brain tumors. Early detection of brain tumors utilizing MRI images and the tumor segmentation method appears promising. However, there is still a long way to go before the tumor site can be precisely recognized and categorized. In order to detect brain tumors in their early stages, We employed a range of MRI brain tumor images in our investigation. Deep learning models have also an enormous contribution to classification and detection. We developed a ResNet101 model for the early identification of brain cancers, and utilizing a large number of MRI images, we found encouraging results. We used a range of key performance indicators to verify the DL models effectiveness.During the evaluation process, models are used. In addition to the suggested model, we also took into consideration to evaluate our results, we used a few additional DL models. Because the ResNet101 had numerous layers and the computer required an effective GPU. The training procedure was lengthy. If the dataset is huge, such as a thousand rows, images, training would take longer. We reduced the size of our GPU system after improving the training time.

9.2 Future Enhancement

Future enhancements in brain tumor classification using Convolutional Neural Networks (CNN) may include:

1. Multi-modal Integration: Incorporating various imaging modalities like MRI, CT scans, and functional imaging data to improve classification accuracy.
2. Transfer Learning: Leveraging pre-trained models on large datasets to enhance the performance of brain tumor classification, especially when dealing with limited labeled data.
3. Explainability and Interpretability: Developing methods to interpret and explain CNN decisions, providing insights into the features contributing to tumor classification for better clinical acceptance.

4. Real-time Processing: Implementing real-time processing for quicker and more efficient diagnosis, which is crucial in medical scenarios.
5. Integration with Clinical Data: Combining CNN outputs with relevant clinical information to create a comprehensive diagnostic tool that considers both imaging and patient-specific data.

CHAPTER- X

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INTERNAL QUALITY ASSURANCE CELL PROJECT AUDIT REPORT

This is to certify that the project work entitled "TITLE OF THE PROJECT" categorized as an internal project done by STUDENT NAME 1, STUDENT NAME 2, STUDENT NAME 3 of the Department of Computer Science and Engineering, under the guidance of GUIDE NAME during the Even semester of the academic year 2023 - 2024 are as per the quality guidelines specified by IQAC.

Quality Grade

Deputy Dean (IQAC)

Administrative Quality Assurance **Dean (IQAC)**